CHAPTER

15 Communication Systems

1. Bandwidth of video signals is 4.2 MHz.
2. Coaxial cable is a widely used wire medium, which offers a bandwidth of approximately 750 MHz.
3. For transmission over long distances, signals are radiated into space using devices called antennas. The radiated signals propagate as electromagnetic waves and the mode of propagation is influenced by the presence of the earth and its atmosphere. Near the surface of the earth, electromagnetic waves propagate as surface waves. Surface wave propagation is useful up to a few MHz frequencies.
4. Communication through free space using radio waves takes place over a very wide range of frequencies from a few hundreds of KHz to a few GHz.
5. To radiate signals with high efficiency, the antennas should have a size comparable to the wavelength λ of the signal (at least ~ $\lambda/4$)
6. If an antenna radiates electromagnetic waves from a height h, then the range d is given by $\sqrt{2Rh}$ where R is the radius of earth.
7. The maximum line of sight distance d_M between the two antennas having heights h_T and h_R above the earth is given by $d_{ü} = \sqrt{2Rh} + \sqrt{2Rh}$
8. Low frequencies cannot be transmitted to long distances. Therefore, they are superimposed on a high frequency carrier signal by a process known as modulation.
9. The method in which the amplitude of carrier wave is varied in accordance with the modulating signal keeping the frequency and phase of carrier wave constant is called amplitude modulation (AM).
10. Amplitude modulated signal contains frequencies $(\omega_c - \omega_m)$, ω_c and $(\omega_c + \omega_m)$.
11. Amplitude modulated waves can be produced by application of the message signal and the carrier wave to a non-linear device, followed by a band pass filter.
12. AM detection, which is the process of recovering the modulating signal from an AM waveform, is carried out using a rectifier and an envelope detector.